

SultiAI

Database Schema

Technical documentation of the SultiAI learning platform

Version 1.0 · Generated 18 September 2026

41 tables · 4 database dialects

PostgreSQL (Supabase) · SQLite (dev default) · MySQL (optional) · MongoDB (optional)
powers civic language preservation, AI tutor, adaptive learning, and gamification

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1 Overview

This document describes the database schema of the SultiAI — a bilingual Cebuano–English (Bisaya–English) learning platform. It catalogs every table, column, constraint, index, seed set, and security policy. The production schema is PostgreSQL (Supabase); the default local runtime uses SQLite via Drizzle ORM, with optional MySQL and MongoDB backends.

Table inventory

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- Gamification & XP — xp_logs, user_achievements, user_badges, completed_challenges, ai_recommendations
- System & audit — user_sessions, audit_logs, saved_phrases, feedback, platform_settings

2 Technology Stack & Environments

Three relational dialects plus one document store are supported. The canonical DDL lives in `server/src/db/connection.ts` and in the Supabase migrations under `supabase/migrations/`.

Environment	Engine	Notes
Local / dev (default)	SQLite via better-sqlite3	<code>server/.env</code> has no <code>DB_DIALECT</code> != <code>sqlite</code>
Production	PostgreSQL 15 (Supabase)	41 tables + RLS policies; migrations in <code>supabase/migrations/</code>
Optional	MySQL 8 (mysql2)	Drizzle <code>mysql</code> schema + init script
Optional	MongoDB (mongoose)	<code>TutorSession</code> & <code>LearnerProfile</code> mirrors if <code>MONGODB_URI</code> set
Caching / rate limiting	Upstash Redis	no persistent schema

3 Conventions & Type Mapping

- Primary keys — serial auto-increments (users, sessions, posts, posts_id, ...). Distributed/filtered by table uses TEXT keys (id).
- Timestamps — PostgreSQL uses

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4 Dialect Coverage Matrix

Availability across the supported relational dialects. PG = Supabase PostgreSQL (production), SQLite = default dev runtime, MySQL = optional dialect.

Table	PG	SQLite	MySQL
avatars	•	•	•
users	•	•	•
roles	•	•	—
permissions	•	•	—
role_permissions	•	•	—
user_settings	•	•	•
notification_preferences	•	•	•
notifications	•	•	•
speech_records	•	•	•
translations	•	•	•
phrase_recommendations	•	•	•
learning_modules	•	•	•
learning_progress	•	•	•
learner_profiles	•	•	•
learning_analytics	•	•	•
daily_activity	•	•	•
conversations	•	•	•
conversation_messages	•	•	•
conversation_summaries	•	•	•
tutor_sessions	•	•	•

vocabulary_reviews	'	'	'
pronunciation_attempts	'	'	'
preserved_words	'	'	'
verifications	'	'	'
verification_requests	'	'	'
community_posts	'	'	'
comments	'	'	'
community_reports	'	'	'
bookmarks	'	'	'
likes	'	'	'
follows	'	'	'
xp_logs	'	'	'
user_achievements	'	'	'
user_badges	'	'	'
completed_challenges	'	'	'
ai_recommendations	'	'	'
user_sessions	'	'	'
audit_logs	'	'	'
saved_phrases	'	'	'
feedback	'	'	'
platform_settings	'	—	—

Optional MongoDB: TutorSession and LearnerProfile are mirrored document models consulted only when MONGODB_URI is configured.

2 Identity, Authentication & Access Control

2.1 avatars

Avatar catalog referenced by users.avatar_id.

Column	Type	Constraints / Notes
avatar_id	SERIAL	PRIMARY KEY
avatar_name	TEXT	NOT NULL
avatar_image	TEXT	NOT NULL · URL

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2.2 users

Application users. Extends Supabase auth.users — one row per registered learner.

Column	Type	Constraints / Notes
user_id	SERIAL	PRIMARY KEY
fullname	TEXT	NOT NULL
username	TEXT	
email	TEXT	NOT NULL · UNIQUE
password_hash	TEXT	NOT NULL
clerk_id	TEXT	Clerk managed auth
google_id	TEXT	OAuth (Google)
supabase_id	TEXT	UNIQUE
avatar_id	INTEGER	FK !' avatars(avatar_id) · DEFAULT 1
preferred_lang	TEXT	DEFAULT 'English'
learning_lang	TEXT	DEFAULT 'Bisaya'
country	TEXT	
role	TEXT	NOT NULL · DEFAULT 'user' ('user' 'moderator' 'admin')
status	TEXT	NOT NULL · DEFAULT 'approved'
is_verified	INTEGER	DEFAULT 0 (boolean)
created_at	TIMESTAMPTZ	DEFAULT NOW()
role_id	INTEGER	FK !' roles(role_id) · added by RBAC migration

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2.3 roles

RBAC roles. Seeded with user, moderator, admin.

Column	Type	Constraints / Notes
role_id	SERIAL	PRIMARY KEY
name	TEXT	NOT NULL · UNIQUE
description	TEXT	
created_at	TIMESTAMPTZ	DEFAULT NOW()

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2.4 permissions

Granular action/resource permission catalog. Seeded with 25 permissions.

Column	Type	Constraints / Notes
permission_id	SERIAL	PRIMARY KEY
name	TEXT	NOT NULL · UNIQUE · e.g. posts:moderate
resource	TEXT	NOT NULL · users posts comments verifications lessons ai achievements settings feedback audit
action	TEXT	NOT NULL · read write delete moderate approve manage_roles manage_status use resolve
description	TEXT	

2.5 role_permissions

Junction table mapping roles to permissions (many-to-many).

Column	Type	Constraints / Notes
role_id	INTEGER	NOT NULL · FK !' roles(role_id) ON DELETE CASCADE · part of composite PK
permission_id	INTEGER	NOT NULL · FK !' permissions(permission_id) ON DELETE CASCADE · part of composite PK

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3 User Preferences & Notifications

3.1 user_settings

1:1 with users — per-user UI/speech preferences.

Column	Type	Constraints / Notes
setting_id	SERIAL	PRIMARY KEY
user_id	INTEGER	NOT NULL · UNIQUE · FK !' users(user_id) ON DELETE CASCADE
dark_mode	INTEGER	DEFAULT 0 (boolean)
speech_speed	REAL	DEFAULT 1.0
voice_gender	TEXT	DEFAULT 'neutral'

3.2 notification_preferences

1:1 with users — opt-in/out flags and times for notifications.

Column	Type	Constraints / Notes
id	SERIAL	PRIMARY KEY
user_id	INTEGER	NOT NULL · UNIQUE · FK !' users(user_id) ON DELETE CASCADE
daily_reminder	INTEGER	DEFAULT 1
daily_reminder_hour	INTEGER	DEFAULT 9
daily_reminder_minute	INTEGER	DEFAULT 0
streak_reminder	INTEGER	DEFAULT 1
review_reminder	INTEGER	DEFAULT 1
weekly_report	INTEGER	DEFAULT 1
achievement_alerts	INTEGER	DEFAULT 1
community_alerts	INTEGER	DEFAULT 1
updated_at	TIMESTAMPTZ	DEFAULT NOW() · updated by trigger

3.3 notifications

In-app notification inbox entries.

Column	Type	Constraints / Notes
notify_id	SERIAL	PRIMARY KEY
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE

title	TEXT	
message	TEXT	
is_read	INTEGER	DEFAULT 0
created_at	TIMESTAMPTZ	DEFAULT NOW()

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Note:
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types
(constant
values):
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4 Speech Recognition & Translation

4.1 speech_records

Raw speech-to-text practice attempts per user.

Column	Type	Constraints / Notes
speech_id	SERIAL	PRIMARY KEY
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
audio_path	TEXT	
recognized_text	TEXT	
language_detected	TEXT	
confidence	REAL	DEFAULT 0
created_at	TIMESTAMPTZ	DEFAULT NOW()

4.2 translations

Translation output tied to a speech record.

Column	Type	Constraints / Notes
translation_id	SERIAL	PRIMARY KEY
speech_id	INTEGER	NOT NULL · FK !' speech_records(speech_id) ON DELETE CASCADE
source_language	TEXT	
target_language	TEXT	
translated_text	TEXT	

4.3 phrase_recommendations

AI-suggested phrases derived from a speech record.

Column	Type	Constraints / Notes
recommendation_id	SERIAL	PRIMARY KEY
speech_id	INTEGER	NOT NULL · FK !' speech_records(speech_id) ON DELETE CASCADE
recommended_phrase	TEXT	
intent	TEXT	
confidence	REAL	DEFAULT 0

5 Learning & Progress

5.1 learning_modules

Curated lessons/modules catalog.

Column	Type	Constraints / Notes
module_id	SERIAL	PRIMARY KEY
module_title	TEXT	NOT NULL
difficulty	TEXT	DEFAULT 'beginner'
language	TEXT	
created_at	TIMESTAMPTZ	DEFAULT NOW()

5.2 learning_progress

Per-user completion state of a learning module.

Column	Type	Constraints / Notes
progress_id	SERIAL	PRIMARY KEY
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
module_id	INTEGER	NOT NULL · FK !' learning_modules(module_id) ON DELETE CASCADE
completion_percent	REAL	DEFAULT 0
created_at	TIMESTAMPTZ	DEFAULT NOW()

5.3 learner_profiles

1:1 with users — aggregate learner stats used by the adaptive engine.

Column	Type	Constraints / Notes
profile_id	SERIAL	PRIMARY KEY
user_id	INTEGER	NOT NULL · UNIQUE · FK !' users(user_id) ON DELETE CASCADE
level	TEXT	DEFAULT 'beginner'
strengths	TEXT	
weak_areas	TEXT	
common_mistakes	TEXT	
total_xp	INTEGER	DEFAULT 0
coins	INTEGER	DEFAULT 0
streak	INTEGER	DEFAULT 0
daily_xp	INTEGER	DEFAULT 0
daily_goal	INTEGER	DEFAULT 50
total_sessions	INTEGER	DEFAULT 0
last_active	TEXT	

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5.4 learning_analytics

1:1 with users — computed learning telemetry for dashboards.

Column	Type	Constraints / Notes
id	SERIAL	PRIMARY KEY
user_id	INTEGER	NOT NULL · UNIQUE · FK !' users(user_id) ON DELETE CASCADE
totalSpeakingSeconds	INTEGER	DEFAULT 0
totalWordsLearned	INTEGER	DEFAULT 0
totalPronunciationAttempts	INTEGER	DEFAULT 0
avgPronunciationAccuracy	REAL	DEFAULT 0
avgSessionDuration	REAL	DEFAULT 0
favoriteCategory	TEXT	
weakestCategory	TEXT	
weeklyXp	TEXT	DEFAULT '[]' (JSON array)
lastCalculated	TIMESTAMPTZ	

5.5 daily_activity

Per-day XP totals for streak & goal tracking.

Column	Type	Constraints / Notes
activity_id	SERIAL	PRIMARY KEY
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
activity_date	TEXT	NOT NULL (YYYY-MM-DD)
xp_earned	INTEGER	DEFAULT 0

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6 AI Tutor & Conversations

6.1 conversations

Top-level AI tutor chat threads.

Column	Type	Constraints / Notes
conversation_id	SERIAL	PRIMARY KEY
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
title	TEXT	
created_at	TIMESTAMPTZ	DEFAULT NOW()

6.2 conversation_messages

Messages within a conversation.

Column	Type	Constraints / Notes
message_id	SERIAL	PRIMARY KEY
conversation_id	INTEGER	NOT NULL · FK !' conversations(conversation_id) ON DELETE CASCADE
sender	TEXT	NOT NULL · DEFAULT 'user'
message	TEXT	
translated_message	TEXT	
created_at	TIMESTAMPTZ	DEFAULT NOW()

6.3 conversation_summaries

Periodic memory snapshots of tutor conversations (long-term memory).

Column	Type	Constraints / Notes
id	TEXT	PRIMARY KEY
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
summary	TEXT	NOT NULL

topics	TEXT	DEFAULT '[]'
vocabulary_learned	TEXT	DEFAULT '[]'
duration	INTEGER	DEFAULT 0 (seconds)
timestamp	TIMESTAMPTZ	DEFAULT NOW()

6.4 tutor_sessions

AI tutor sessions (voice + chat). messages stores a JSON array of exchanges.

Column	Type	Constraints / Notes
session_id	SERIAL	PRIMARY KEY
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
messages	TEXT	JSON array of {role, content}
summary	TEXT	
started_at	TIMESTAMPTZ	DEFAULT NOW()
ended_at	TIMESTAMPTZ	
xp_earned	INTEGER	DEFAULT 0

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Note:
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URI is set.

7 Vocabulary, Pronunciation & Preservation

7.1 vocabulary_reviews

Spaced-repetition (SM-2) vocabulary cards.

Column	Type	Constraints / Notes
id	TEXT	PRIMARY KEY
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
word	TEXT	NOT NULL
translation	TEXT	DEFAULT ''
pronunciation	TEXT	DEFAULT ''

ipa	TEXT	
category	TEXT	DEFAULT 'custom'
difficulty	INTEGER	DEFAULT 1 (1–5)
mastery	REAL	DEFAULT 0 (0–1)
review_count	INTEGER	DEFAULT 0
ease_factor	REAL	DEFAULT 2.5 (SM-2)
interval	INTEGER	DEFAULT 1 (days)
next_review	TEXT	NOT NULL (due date)
last_review	TEXT	
is_favorite	INTEGER	DEFAULT 0
usage_frequency	INTEGER	DEFAULT 0
created_at	TIMESTAMPTZ	DEFAULT NOW()
updated_at	TIMESTAMPTZ	DEFAULT NOW() · updated by trigger

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7.2 pronunciation_attempts

Per-word pronunciation scoring events.

Column	Type	Constraints / Notes
id	TEXT	PRIMARY KEY
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
word	TEXT	NOT NULL
phonetic_expected	TEXT	DEFAULT ''
phonetic_heard	TEXT	DEFAULT ''
accuracy	REAL	DEFAULT 0
confidence	REAL	DEFAULT 0
mistakes	TEXT	DEFAULT '[]' (JSON phoneme mistakes)
lesson_context	TEXT	
timestamp	TIMESTAMPTZ	DEFAULT NOW()

7.3 preserved_words

Crowd-sourced Bisaya word preservation entries.

Column	Type	Constraints / Notes
word_id	SERIAL	PRIMARY KEY
word	TEXT	NOT NULL
definition	TEXT	
part_of_speech	TEXT	
dialectal_region	TEXT	
bisaya_example	TEXT	
english_example	TEXT	
pronunciation_guide	TEXT	
submitted_by	INTEGER	FK !' users(user_id) ON DELETE SET NULL
source	TEXT	DEFAULT 'learner'
status	TEXT	DEFAULT 'pending' (pending approved rejected)
verification_count	INTEGER	DEFAULT 0
created_at	TIMESTAMPTZ	DEFAULT NOW()

7.4 verifications

Pronunciation verification workflow rows.

Column	Type	Constraints / Notes
id	SERIAL	PRIMARY KEY
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
verified_by	INTEGER	FK !' users(user_id) ON DELETE SET NULL

status	TEXT	DEFAULT 'pending'
created_at	TIMESTAMPTZ	DEFAULT NOW()

7.5 verification_requests

Requests to verify a preserved word pronunciation.

Column	Type	Constraints / Notes
request_id	SERIAL	PRIMARY KEY
word_id	INTEGER	FK !' preserved_words(word_id) ON DELETE CASCADE
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
audio_path	TEXT	
recorded_text	TEXT	
verifier_id	INTEGER	FK !' users(user_id) ON DELETE SET NULL
score	REAL	
feedback	TEXT	
status	TEXT	DEFAULT 'pending'
created_at	TIMESTAMPTZ	DEFAULT NOW()
verified_at	TIMESTAMPTZ	

8 Community

8.1 community_posts

Community forum posts (can carry a phrase + translation).

Column	Type	Constraints / Notes
post_id	SERIAL	PRIMARY KEY
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
title	TEXT	
content	TEXT	
phrase	TEXT	
translation	TEXT	
category	TEXT	
likes_count	INTEGER	DEFAULT 0 (denormalized)
bookmarks_count	INTEGER	DEFAULT 0 (denormalized)
is_featured	INTEGER	DEFAULT 0
created_at	TIMESTAMPTZ	DEFAULT NOW()

8.2 comments

Comments on community posts.

Column	Type	Constraints / Notes
comment_id	SERIAL	PRIMARY KEY

post_id	INTEGER	NOT NULL · FK !' community_posts(post_id) ON DELETE CASCADE
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
comment	TEXT	
created_at	TIMESTAMPTZ	DEFAULT NOW()

8.3 community_reports

Moderation reports against posts.

Column	Type	Constraints / Notes
report_id	SERIAL	PRIMARY KEY
post_id	INTEGER	NOT NULL · FK !' community_posts(post_id) ON DELETE CASCADE
reporter_id	INTEGER	FK !' users(user_id) ON DELETE SET NULL
reason	TEXT	
status	TEXT	DEFAULT 'open' (open resolved dismissed)
created_at	TIMESTAMPTZ	DEFAULT NOW()

8.4 bookmarks

User bookmarking of posts.

Column	Type	Constraints / Notes
id	SERIAL	PRIMARY KEY
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
post_id	INTEGER	NOT NULL · FK !' community_posts(post_id) ON DELETE CASCADE
created_at	TIMESTAMPTZ	DEFAULT NOW()

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Note:
UNIQUE(user_id,
post_id)
—
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post.

8.5 likes

User likes on posts.

Column	Type	Constraints / Notes
id	SERIAL	PRIMARY KEY
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
post_id	INTEGER	NOT NULL · FK !' community_posts(post_id) ON DELETE CASCADE
created_at	TIMESTAMPTZ	DEFAULT NOW()

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Note:
UNIQUE(user_id,
post_id)
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8.6 follows

User-to-user follow graph.

Column	Type	Constraints / Notes
id	SERIAL	PRIMARY KEY
follower_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
following_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
created_at	TIMESTAMPTZ	DEFAULT NOW()

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Note:
UNIQUE(follower_id,
following_id).
id).

9 Gamification & XP

9.1 xp_logs

Append-only ledger of XP earned, per source.

Column	Type	Constraints / Notes
id	TEXT	PRIMARY KEY
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
amount	INTEGER	NOT NULL
source	TEXT	NOT NULL · one of XP_VALUES keys
description	TEXT	
timestamp	TIMESTAMPTZ	DEFAULT NOW()

9.2 user_achievements

Achievements unlocked by a user.

Column	Type	Constraints / Notes
user_achievement_id	SERIAL	PRIMARY KEY
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
achievement_id	TEXT	NOT NULL · e.g. first_100_xp

unlocked_at	TIMESTAMPTZ
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Note:
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9.3 user_badges

Badges earned by a user.

Column	Type	Constraints / Notes
user_badge_id	SERIAL	PRIMARY KEY
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
badge_id	TEXT	NOT NULL
earned_at	TIMESTAMPTZ	

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Note:
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9.4 completed_challenges

Challenges completed by a user.

Column	Type	Constraints / Notes
id	SERIAL	PRIMARY KEY
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
challenge_id	TEXT	NOT NULL
completed_at	TIMESTAMPTZ	DEFAULT NOW()

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9.5 ai_recommendations

Personalized AI recommendations (lessons, phrases, drills).

Column	Type	Constraints / Notes
id	TEXT	PRIMARY KEY
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
recommendation_type	TEXT	NOT NULL
content	TEXT	NOT NULL
priority	INTEGER	DEFAULT 0
is_applied	INTEGER	DEFAULT 0
created_at	TIMESTAMPTZ	DEFAULT NOW()
applied_at	TIMESTAMPTZ	

10 System & Audit

10.1 user_sessions

Active auth sessions / refresh tokens.

Column	Type	Constraints / Notes
id	TEXT	PRIMARY KEY
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
refresh_token	TEXT	NOT NULL
device_info	TEXT	
ip_address	TEXT	
expires_at	TEXT	NOT NULL
created_at	TIMESTAMPTZ	DEFAULT NOW()

10.2 audit_logs

Immutable audit trail for admin/security events.

Column	Type	Constraints / Notes
id	SERIAL	PRIMARY KEY
user_id	INTEGER	no FK — may reference a deleted actor
action	TEXT	NOT NULL
resource_type	TEXT	

resource_id	TEXT	
details	TEXT	
ip_address	TEXT	
timestamp	TIMESTAMPTZ	DEFAULT NOW()

10.3 saved_phrases

Phrases saved by a learner for review.

Column	Type	Constraints / Notes
phrase_id	SERIAL	PRIMARY KEY
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
phrase	TEXT	NOT NULL
language	TEXT	
category	TEXT	
created_at	TIMESTAMPTZ	DEFAULT NOW()

10.4 feedback

In-app user feedback ratings.

Column	Type	Constraints / Notes
feedback_id	SERIAL	PRIMARY KEY
user_id	INTEGER	NOT NULL · FK !' users(user_id) ON DELETE CASCADE
functionality	INTEGER	DEFAULT 0 (rating)
usability	INTEGER	DEFAULT 0 (rating)
reliability	INTEGER	DEFAULT 0 (rating)
resolved	INTEGER	DEFAULT 0
created_at	TIMESTAMPTZ	DEFAULT NOW()

10.5 platform_settings

Global key/value store (JSONB), scoped to platform administration.

Column	Type	Constraints / Notes
key	TEXT	PRIMARY KEY
value	JSONB	NOT NULL
updated_at	TIMESTAMPTZ	DEFAULT NOW() · updated by trigger

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11 Security & Row-Level Security

Every table has Row-Level Security (RLS) enabled on Supabase. Baseline policies grant admins full access; a later migration (20260904_fix_rls_policies.sql) replaces the broad USING (true) admin policies with role- and ownership-aware policies.

RLS helper functions

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Service-role clients (e.g. the Node backend) bypass RLS entirely.

Timestamps & triggers

update_updated_at() is a PL/pgSQL trigger that sets updated_at = NOW() on BEFORE UPDATE. Applied to user_settings, vocabulary_reviews, notification_preferences, and platform_settings.

12 Seed Data & Domain Enumerations

Reusable constants and seeded reference data consumed by the application. Source: server/src/config/constants.ts and the RBAC migration.

XP values (xp_logs.source)

Each XP event is attributed to a source key. xp_logs.source must be one of:

VOICE_PRACTICE_LEARN	25 XP	VOICE_PRACTICE_TURN	15 XP
CHAT_MESSAGE	10 XP	LESSON_GENERATE	20 XP
LESSON_COMPLETE	30 XP	ROLEPLAY_START	15 XP
FLASHCARD_KNOWN	10 XP	FLASHCARD_REVIEW	5 XP
WHISPER_INTERACTION	5 XP	AR_SCAN	5 XP
SAVE_PHRASE	2 XP	ACHIEVEMENT_BONUS	100 XP
DAILY_REWARD	50 XP	DAILY_GOAL_COMPLETE	50 XP
TUTOR_SESSION_COMPLETE	10 XP	TUTOR_CHAT	10 XP
TUTOR_LESSON	20 XP		

Vocabulary categories (vocabulary_reviews.category)

greetings · food · family · travel · numbers · time · emotions · body · nature · work · culture · custom

Notification types

daily_reminder · streak_reminder · review_vocabulary · weekly_report · new_challenge · achievement_unlocked · community_reply · goal_achieved

Level thresholds (LEVEL_THRESHOLDS)

Level	XP	Label
1	0	Starter

2	100	Beginner
3	500	Intermediate
4	2000	Advanced
5	5000	Native-like

Achievements (seeded IDs)

first_100_xp · thousand_xp · five_thousand_xp · streak_3 · streak_7 · streak_30 · daily_goal

RBAC seed data

Roles: user, moderator, admin. 25 permissions across 10 resources (users, posts, comments, verifications, lessons, ai, achievements, settings, feedback, audit).

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